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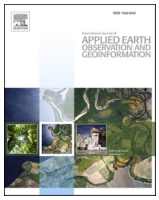

Non-negative matrix factorization-assisted independent component analysis for land subsidence feature decomposition and attribution in Lubei Plain, and further integration with deep learning for subsidence prediction

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ABSTRACT

Interferometric Synthetic Aperture Radar (InSAR) has become crucial for large-scale land subsidence monitoring, but its results represent complex superpositions of deformation signals from multiple compressible layers driven by various anthropogenic and natural factors. Decomposing subsidence patterns by different driving factors is vital for attribution and prediction. However, existing feature decomposition methods suffer from problems of incomplete decomposition and difficulties in direction determination. This study employs non-negative matrix factorization (NMF) assisted independent component analysis (ICA) to achieve feature decomposition and attribution of land subsidence in the Lubei Plain (LBP) from 2018 to 2023, and integration with deep learning to realize land subsidence prediction. Key contributions include: (1) Using NMF component directions as reference to determine three independent component directions for subsequent attribution and prediction analysis. (2) Groundwater wells in high-spatial-score regions were selected to analyze the relationship between ICs, deformation, and groundwater levels. IC1 showed a high correlation exceeding 0.8 with subsidence, reflecting near-linear patterns from deep confined aquifer over-exploitation and petroleum extraction. IC2 reached maximum correlations of 0.72 with subsidence and 0.56 with groundwater levels (GWL), reflecting inter-annual precipitation-driven variations. Similarly, IC3 exhibited maximum correlations of 0.63 and 0.71, respectively, characterizing seasonal cycles linked to agricultural irrigation and precipitation. (3) The proposed prediction model demonstrated excellent performance with RMSE of 0.06–0.69 and R^2 of 0.80–0.86 in test datasets. This study provides a new reference perspective for future driving factor analysis and regulation of land subsidence in the Lubei Plain, and offers a new method for land subsidence prediction.

1. Introduction

Land subsidence, as a common geological phenomenon, poses significant threats to urban development and safety (Bagheri-Gavkosh et al., 2021; Chen et al., 2020; Riel et al., 2018; Zhao et al., 2019). Interferometric Synthetic Aperture Radar (InSAR) has evolved into an effective land subsidence observation technique (Biggs and Wright, 2020; Lai et al., 2024; Massonnet and Feigl, 1998), characterizing the spatiotemporal evolution of surface deformation through continuous radar imagery and possessing unparalleled large-scale, high-precision deformation monitoring capabilities compared to traditional techniques. Common approaches include Persistent Scatterer Interferometric Synthetic Aperture Radar (PS-InSAR) and Small Baseline Subset Interferometric Synthetic Aperture Radar (SBAS-InSAR). PS-InSAR is a remote sensing technique that utilizes persistent scatterers in long time-series radar images to monitor millimeter-level surface deformation with high precision (Ferretti et al., 2002). SBAS-InSAR combines multiple SAR images with short spatiotemporal baselines and employs distributed scatterers to retrieve surface deformation time series (Berardino et al., 2002). Although SBAS-InSAR employs distributed

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targets instead of PS points and typically exhibits lower monitoring accuracy and reliability at PS point locations compared to PS-InSAR, it can acquire more monitoring data points in areas with sparse PS point distribution (such as vegetated areas and rural regions). It should be noted that whether SBAS-InSAR can obtain denser and more accurate deformation monitoring results largely depends on factors such as surface characteristics of the study area, temporal baselines, and atmospheric conditions. However, in areas with low coherence, SBAS-InSAR typically enables deformation monitoring where PS-InSAR may be ineffective due to insufficient persistent scatterers.

Surface deformation information obtained from InSAR contains a mixture of various geophysical factors (Cohen-Waeber et al., 2018), representing the cumulative deformation of different compressible layers caused by various anthropogenic (such as industrial, commercial, and agricultural activities) (Jing et al., 2021; Verberne et al., 2023; Zhang et al., 2021) and natural (such as climate change and crustal movement) driving factors (Guo et al., 2017; Xu et al., 2008). Meanwhile, different driving factors exhibit their own characteristics. Some driving factors show large interannual variations and are significantly influenced by policy factors, such as groundwater extraction and petroleum extraction (Xu et al., 2025; Zhang et al., 2021). Others exhibit clear periodic characteristics with relatively small interannual changes, such as subsidence variations of aquifers influenced by various climatic factors (such as temperature and precipitation)(Guzy et al., 2025; Wang et al., 2024). Furthermore, the characteristics of the aquifer system should also be considered. The subsidence and deformation characteristics of the aquifer system are mainly controlled by the magnitude of effective stress increment caused by pore pressure decline, the relationship between current effective stress and preconsolidation stress, and the geotechnical composition of aquifer-aquitard systems, etc. (Gambolati and Teatini, 2015; Guzy and Malinowska, 2020). During groundwater extraction that causes water head decline, different types of aquifers exhibit varying response mechanisms. For aquifers dominated by sandy soils, due to their high permeability and relatively low compressibility, elastic deformation primarily occurs under pore water pressure changes, and land subsidence rebounds relatively promptly when groundwater levels recover (Wang, 2023). For aquifers dominated by clayey soils, due to their low permeability and high compressibility characteristics, pore water pressure decline induces plastic deformation of the soil skeleton and particle rearrangement, making subsidence less responsive to timely recovery when groundwater levels rise (Jiang et al., 2018; Wang et al., 2024).

Due to the differences in the aforementioned driving factors and aquifer characteristics, different subsidence time series exhibit non-Gaussian independence. This provides a foundation for extracting independent subsidence patterns from mixed subsidence data, thereby enabling better attribution, analysis, and prediction of the land subsidence patterns represented by the independent components. Some scholars have used Principal Component Analysis (PCA) to decompose land subsidence data obtained from InSAR observations (Festa et al., 2023; Peng et al., 2022). Although PCA captures the main characteristics of the original signal, each PC still contains contributions from all signals (Peng, 2023). Independent Component Analysis (ICA) achieves signal decomposition based on the non-Gaussian independence of signals, originating from the original blind source signal separation (BSS). Previous studies have used ICA to decompose surface deformation data observed by InSAR and GNSS (Ebmeier, 2016; Frappart et al., 2011; Lai et al., 2024; Peng et al., 2022; Wang et al., 2025) to discover their internal characteristics. Since the independent components obtained by ICA decomposition are dimensionless and randomly oriented, this study employs Non-negative Matrix Factorization (NMF) to extract regional main subsidence patterns to determine the direction of independent components. Scholars have used NMF to decompose land subsidence data to obtain the main subsidence characteristics within regions (Brunet et al., 2004; Gu et al., 2023). The components and coefficients of NMF are all positive values, avoiding negative value cancellation effects, ensuring directional correctness, and providing reliable reference for determining the direction and existence of independent components (Lee and Seung, 2000).

Meanwhile, scholars have combined ICA with Long Short-Term Memory (LSTM) neural networks for subsidence prediction and achieved good results (Peng et al., 2024). LSTM, as a neural network model commonly used for time series prediction, is a variant of traditional Recurrent Neural Networks (RNN) that addresses gradient vanishing and gradient explosion problems caused by long-term dependencies through the introduction of gating mechanisms (Kumar et al., 2022; Liu and Zhang, 2023). Further improved models such as CNN-LSTM-Attention that combines Convolutional Neural Networks (CNN) with channel attention mechanisms (Hu et al., 2018), which can focus on important temporal dependencies during time series prediction. Considering the characteristics of ICA, signals can be reconstructed through fewer independent components and decomposition matrices, similar to filters that remove noise (Peng et al., 2024; Wang et al., 2025). If simpler independent components can be predicted separately and then reconstructed back to the original signal using decomposition matrices, regional land subsidence prediction can be achieved for the entire area.

The Lubei Plain (LBP) is located in the southeastern part of the North China Plain (NCP) and has experienced large-scale groundwater extraction since the 1970 s. In western cities of the Lubei Plain, such as Dezhou and Liaocheng, deep groundwater has been extensively extracted for agricultural irrigation (Wang et al., 2024; Xu et al., 2025). In eastern cities of the Lubei Plain, such as Dongying, due to the widespread distribution of underground brine, groundwater has been extensively extracted for salt production and aquaculture (Higgins et al., 2013; Zhang et al., 2021). Meanwhile, in Dongying, due to the presence of the Shengli Oilfield, large-scale oil extraction has also caused severe land subsidence (Feng and Xue, 2004).

Regarding the land subsidence phenomenon in the study area, this research utilizes SBAS-InSAR to obtain land subsidence data for the LBP region from 2018 to 2023, employs NMF to extract the main land subsidence characteristics within the region, explores the relationship between NMF-derived components and independent components, and assists in determining the decomposition direction of ICA. Subsequently, driving factor attribution analysis is conducted for the land subsidence patterns represented by different independent components. Finally, combining ICA with CNN-LSTM-Attention models to predict each independent component, and using the decomposition matrix to obtain land subsidence results for the entire region.

2. Study area and data

2.1. Study area

The North China Plain has a temperate monsoon climate, with cold and dry winters and hot and rainy summers (Li et al., 2022; Xu et al., 2025). The annual average precipitation is approximately 670 mm, with rainfall mainly concentrated in July and August. The annual average evaporation ranges between 1000 and 1300 mm, peaking from March to June (Wang et al., 2024).

The geomorphological types are predominantly Yellow River alluvial plains, with coastal plains influenced by marine deposition distributed in the coastal areas (Wang et al., 2012). From the hydrogeological cross-section of Decheng City shown in Fig. 1, it can be observed that aquifers and compressible layers appear alternately within the region, with 8 aquifers and 9 compressible layers identified within a burial depth of 800 m. In the eastern areas such as Dongying and Guangrao, 7 aquifers and 8 compressible layers can be identified within a burial depth of 600 m. Different groundwater utilization methods in various aquifers have led to deformation of different compressible layers.

The primary Land cover type in the region is farmland, mainly cultivating winter wheat and summer corn, implementing a double cropping system with winter wheat and summer corn rotation (Li and

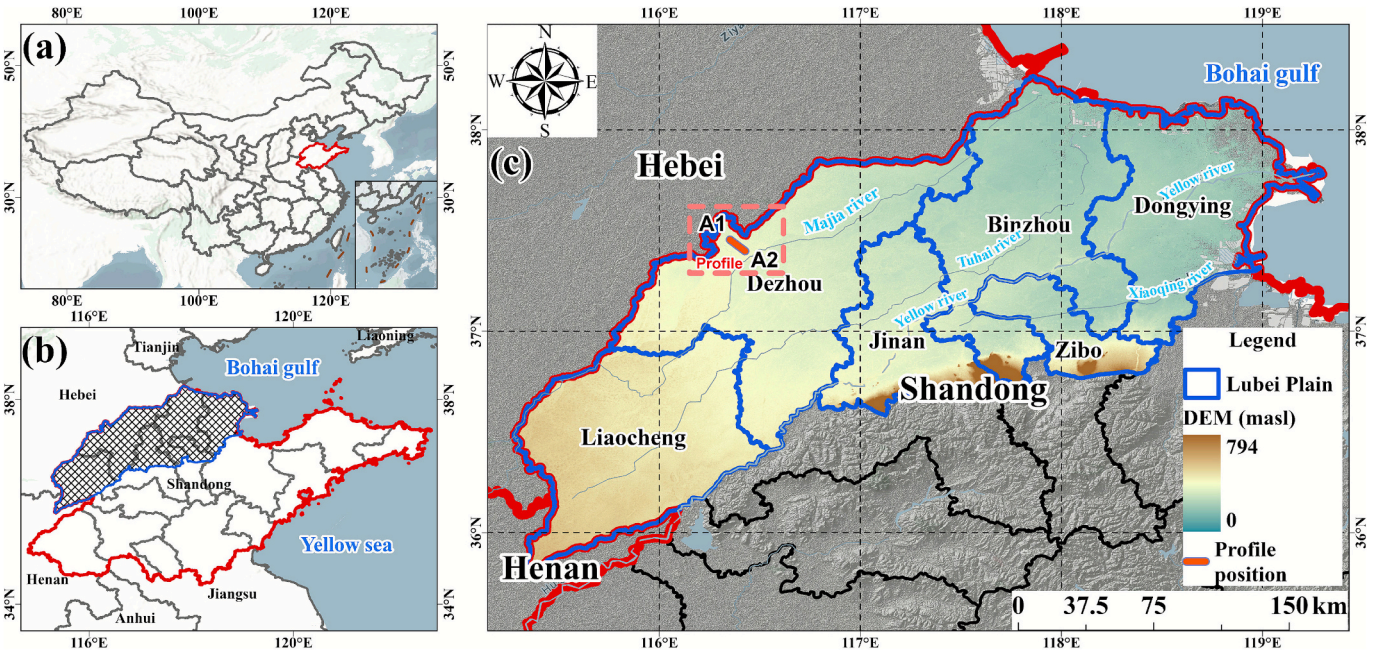

Fig. 1. The location of LBP in China and Shandong Province, as well as the elevation information of the research area. Elevations are given in meters above sea level (masl).

Lei, 2021). In western areas such as Dezhou, deep groundwater extraction is used for agricultural irrigation, with groundwater levels (GWL) having declined by more than 100 m (Hao et al., 2022; Jia et al., 2021). Following the implementation of comprehensive groundwater management policies by the Shandong Provincial Government, GWL began to rebound after 2015, leading to new characteristics in regional water cycles (Di et al., 2020). Within Dongying City at the Yellow River estuary, saline-alkali lands are widely distributed and petroleum resources are abundant, with numerous salt chemical enterprises and petroleum processing companies (Zhang et al., 2019). Fig. 1 (a) shows the location of Shandong Province in China, (b) shows the location of the LBP within Shandong Province, and (c) presents detailed information about the elevation and river systems of the LBP. Fig. 2 shows the hydrogeological profile of the marked area in the study area.

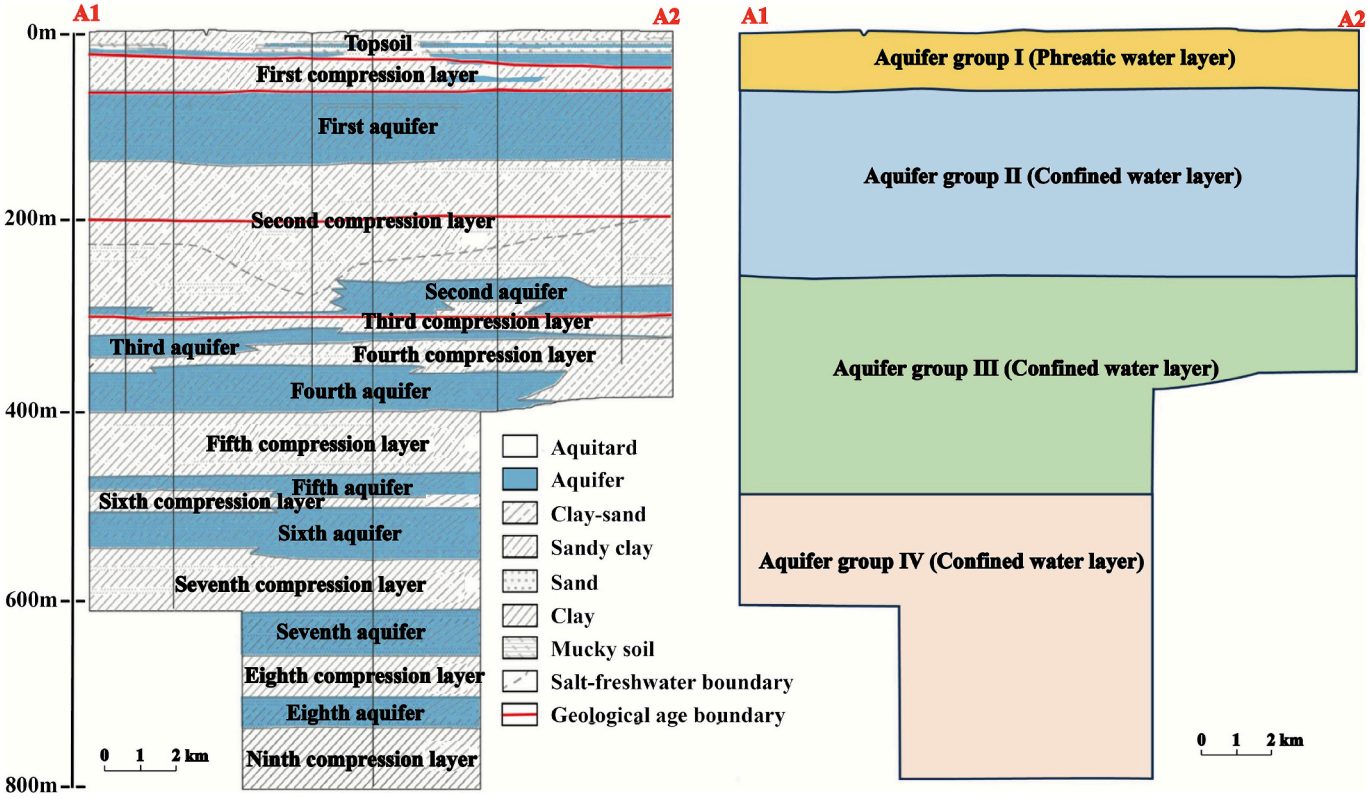

Fig. 2. Hydrogeological profile of a section within the research area(recreated from (Hao et al., 2022)).

2.2 Research data.

2.1.1. Precipitation data

The precipitation data used was the GPM IMERG Final Precipitation L3 data with a spatial resolution of 0.1° and a temporal resolution of 1 month (https://disc.gsfc.nasa.gov).

2.1.2. Land cover data

The Land coverd was based on the China 30-meter Annual Land Cover Dataset (CLCD) (Yang and Huang, 2021)(https://zenodo.org/records/12779975). The data divides Land cover into nine categories: cropland, forest, shurb, grassland, water, snow, barren, impervious and wetland. The data was provided once a year.

2.1.3. Agricultural irrigation water consumption data

The data on farmland irrigation water consumption were obtained from the official website of the Water Conservancy Bureau of each prefecture-level city, and the annual farmland irrigation water consumption data of each county and district in the prefecture-level city were obtained.

2.1.4. Groundwater observation well data

The groundwater observation well data was provided by the China Geological Environment Monitoring Institute. The observation wells were divided into unconfined observation wells and confined groundwater observation wells, and the observation frequency was once a month. There are missing values in the groundwater observation well data. We did not interpolate the data in the subsequent analysis and only used the actual observation data.

2.1.5. Sentinel-1A data

Considering the relatively limited availability of Sentinel-1A descending orbit data in the study area, and the inability to obtain stable data from Sentinel-1B since its malfunction in late 2022, this study opted to exclusively use Sentinel-1A ascending orbit data for subsequent analysis to ensure data consistency and continuity, while avoiding potential impacts from spatial coverage differences between ascending and descending orbit data.

This study used 153 scenes of 69 orbits and 119 frames (69–119), 179 scenes of 142 orbits and 116 frames (142–116) of wide-swath mode (IW) C-band ascending orbit images, all using VV polarization. Sentinel data are available free of charge at the Alaska Satellite Communications Agency (https://search.asf.alaska.edu).

2.1.6. Sentinel-2 data

To further observe the Land cover conditions of subsidence centers with high precision, we filtered Sentinel-2A and Sentinel-2B data in January 2022 using Google Earth Engine (GEE) (https://earthengine.google.com), setting cloud coverage to less than 20 %. Images meeting the criteria were subjected to median composite processing to reduce noise and ensure complete image coverage, followed by true color composite and mosaicking of the results. The revisit period is 5 days with a spatial resolution of 10 m.

3. Methodology

3.1. Time-series InSAR deformation processing

This study uses SBAS-InSAR technology for regional land subsidence monitoring. Compared to PS-InSAR, SBAS-InSAR technology has the advantage of overcoming decorrelation in high decorrelation areas (Crosetto et al., 2016; Shanker et al., 2011). SBAS-InSAR can provide an efficient monitoring solution by constraining temporal and spatial baselines, and can obtain more surface subsidence monitoring points compared to conventional methods. (Du et al., 2020; Figueroa-Miranda et al., 2020; Tang et al., 2020).

Hyp3 (https://hyp3-docs.asf.alaska.edu) is used for interferometric processing, with temporal baseline set to 36 days and spatial baseline set to 200 m to obtain interferometric pairs and perform interferometric processing. After processing, interferograms along with corresponding DEM, look vectors, water masks are simultaneously obtained for subsequent processing.

MintPy (https://github.com/insarlab/MintPy) is used for subsidence time series analysis, using ERA-5 atmospheric reanalysis data from the European Centre for Medium-Range Weather Forecasts to calculate atmospheric delay and subtract it from the interferometric phase. During the time-series InSAR processing using MintPy, the line-of-sight (LOS) deformation measurements will be converted to vertical deformation components based on the satellite incidence angle (Look vectors data acquired from the HyP3 processing stage.).

After converting the land subsidence data from LOS direction to vertical direction, select frame 142–116 as the reference, perform adjustment on the high-coherence overlapping points in the overlap area, calculate correction values to modify the InSAR results of all overlapping points. Finally, the average deformation of all overlapping points, after global offset correction, is calculated to complete the mosaicking of the two frames. The reference equation is as follows, where $V_i(x,y)$ and $V_j(x,y)$ denote the deformation rates of the corresponding high-coherence points within the overlapping region of track i and track j, respectively. ΔV_i and ΔV_j represent the global offsets for track i and track j.

$$\boldsymbol{V_i}(\boldsymbol{x},\boldsymbol{y}) + \Delta \boldsymbol{V_i} = \boldsymbol{V_j}(\boldsymbol{x},\boldsymbol{y}) + \Delta \boldsymbol{V_j} \quad (1)$$

3.2. Principal component analysis

Principal Component Analysis (PCA) (Festa et al., 2023; Maćkiewicz and Ratajczak, 1993) is an important technique for data dimensionality reduction and feature extraction. Its core idea is to project original high-dimensional data into low-dimensional space through linear transformation while maximally preserving the variance information of the data. This study uses PCA to reduce dimensionality and determine the number of independent components to be decomposed. Our strategy for determining the number of independent components is: through PCA dimensionality reduction, using the fewest principal components to retain as much information as possible, and the number of determined principal components is the number of independent component decompositions (Peng et al., 2024).

3.3. Non-negative matrix factorization

Non-negative Matrix Factorization (NMF) is an unsupervised learning algorithm that, similar to Principal Component Analysis (PCA), attempts to express each data point as a weighted sum of several components. A notable characteristic of NMF is that both its components and coefficients are greater than or equal to 0, since the non-negative sum of non-negative components cannot be negative. This also makes the components obtained by NMF more interpretable, as negative components and negative coefficients may lead to cancellation effects (Lee and Seung, 2000).

Therefore, this study uses the component directions obtained from NMF decomposition as reference directions representing the decomposed characteristics of land subsidence, and determines the directions of independent components based on this foundation. That is, for the independent component direction we are uncertain about, the direction of the similar NMF component time series is used as the decomposition direction, while simultaneously providing a reliable basis for its validity and the physical significance it represents.

3.4. Independent component analysis

Independent Component Analysis(ICA) (Lee, 1998) is a statistical

technique used to separate mixed signals. ICA treats observed mixed variables as linear transformations of several statistically independent variables through an unknown matrix. The prerequisites for performing ICA decomposition are: (1) Independent components are statistically independent from each other. (2) Source signals must satisfy non-Gaussian distribution. (3) The number of mixed signals must be greater than or equal to the number of source signals. (4) Noise is controlled within a limited range; in fact, noise can also be identified and removed as an independent component.

The relationship between mixed signals and independent signals can be described by the following equation:

$$\boldsymbol{O}_{t\times n} = \boldsymbol{M}_{t\times l} \times \boldsymbol{S}_{l\times n} \tag{2}$$

Where O is the obtained subsidence sequence matrix, t is the number of observations, and n is the number of spatial points. M is the mixing matrix, which can also be considered as the time series of each independent component, and l is the number of independent components. S is the decomposition matrix of the original matrix, which can also be considered as the spatial scores of independent components, i.e., the contribution degree of each independent component at each point in space. It should be noted that there is uncertainty in the signs of independent components, and the signs can be changed without affecting the model (Peng et al., 2024).

This study uses the FastICA algorithm to perform the above calculations. Taking the FastICA algorithm based on negative entropy as an example, the greater the negative entropy, the greater the non-Gaussianity of the random signal and the stronger the independence, so signals are separated through maximizing non-Gaussianity (Oja and Yuan, 2006). The algorithm steps involve first performing centering and whitening, where centering refers to demeaning the observed data, and whitening refers to decorrelation and scaling.

3.5. Cnn-lstm-attention model

This study uses CNN-LSTM-Attention neural network models to predict each independent component based on ICA decomposition of mixed signals. Considering that ICA decomposition yields subsidence sequences under different subsidence patterns, the prediction difficulty is reduced compared to mixed signals. Assuming that regional subsidence patterns remain relatively stable in the short term, future subsidence can be predicted by utilizing the forecasted independent components through spatial scoring. The model structure diagram is shown in Fig. 3.

CNN extracts time series features after sequence folding through the design of different convolution kernels, and the convolution layer module does not use pooling layers. The attention mechanism module is designed with Squeeze-and-Excitation (SE) channel attention mechanism (Hu et al., 2018), which assigns different weights to different channels through feature compression, feature activation, and channel weight adjustment, that is, assigns different weights to features obtained by different convolution kernels to enhance the feature expression capability. Feature compression is achieved through global average pooling, and its formula is shown in equation (2). H and W represent the height and width of the channel. Z_c represents the global information vector of the c-th channel.

$$\boldsymbol{Z_c} = \frac{1}{\boldsymbol{H}\times\boldsymbol{W}} \sum_{i}^{H} \sum_{j=1}^{W} \boldsymbol{C}(\boldsymbol{i},\boldsymbol{j}) \tag{3}$$

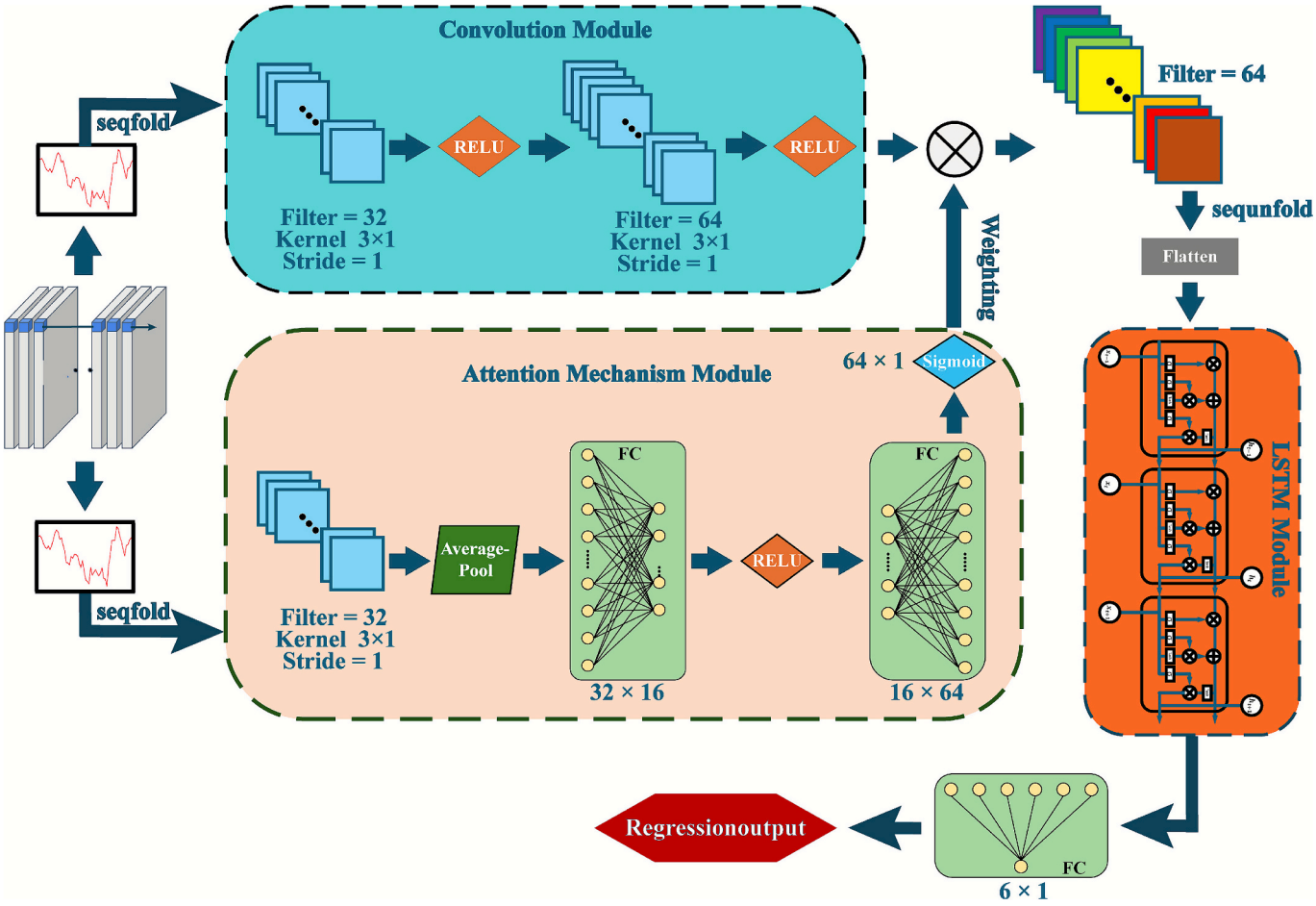

Fig. 3. CNN neural network structure diagram and design parameters.

Feature activation is designed to obtain dependencies between channels, specifically implemented through Sigmoid activation functions. Before the Sigmoid function, two fully connected layers and a ReLU function are designed to parameterize weights, where the parameters of the first fully connected layer are W_1 and the parameters of the second fully connected layer are W_2. The specific implementation formula is shown in equation (3). Where σ represents the Sigmoid activation function, δ represents the ReLU activation function, and S is the learned dependency vector between channels.

$$\boldsymbol{S} = \boldsymbol{\sigma}[\boldsymbol{W}_2\boldsymbol{\delta}(\boldsymbol{W}_1\boldsymbol{Z})] \tag{4}$$

Channel weight adjustment multiplies the learned weight dependency vector with the convolution layer, as shown in equation (4). S_C represents the attention weights obtained from feature activation, U_C represents the channel information processed by the convolution module, and X_C represents the channel information combined with attention weights.

$$\boldsymbol{X}_C = \boldsymbol{S}_C \times \boldsymbol{U}_C \tag{5}$$

The final LSTM module performs prediction on the time series that has been assigned weights and reverse sequence folded. The LSTM model is a variant of the RNN model that solves the gradient explosion and gradient vanishing problems of traditional RNN models by introducing gating mechanisms (forget gate f_t, input gate i_t, and output gate o_t) (Hochreiter and Schmidhuber, 1996; Yu et al., 2019). X_t represents the input at the current time step, h_{t-1} represents the output at the previous time step, and C_{t-1} represents the memory information at the previous time step. Their calculation formulas are shown in equations(5) – (7).

$$\boldsymbol{f}_t = \boldsymbol{\sigma}\left(\boldsymbol{W}_f[\boldsymbol{h}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_f\right) \tag{6}$$

$$\boldsymbol{i}_t = \boldsymbol{\sigma}(\boldsymbol{W}_i[\boldsymbol{h}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_i \tag{7}$$

$$\boldsymbol{o}_t = \boldsymbol{\sigma}(\boldsymbol{W}_i[\boldsymbol{h}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_o) \tag{8}$$

The update formulas for h_t and C_t states are shown in equations (8) – (9), where W and b represent the corresponding weights and biases, respectively.

$$\boldsymbol{C}_t = \boldsymbol{f}_t{*}\boldsymbol{C}_{t-1} + \boldsymbol{i}_t{*}\mathbf{tan}(\boldsymbol{W}_c[\boldsymbol{h}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_c) \tag{9}$$

$$\boldsymbol{h}_t = \boldsymbol{o}_t{*}\mathbf{tan}(\boldsymbol{C}_t) \tag{10}$$

During model training, the Adam gradient descent algorithm is used, with 70 % of the data as the training set and 30 % as the test set. The defined hyperparameters are shown in Table 1. The model is programmed in MATLAB and does not require GPU for parallel computation.

3.6. Evaluation indicators

This study uses the Pearson correlation coefficient r to characterize the correlation between independent components and actual subsidence amounts and observation well GWL (Benesty et al., 2009), and the calculation formula for r is shown in equation (10).

Table 1
Model definition hyperparameters and training strategies.

Model definition hyperparameters and training strategies	Specific details
Initial learning rate	0.001
Max epochs	500
Learning rate drop factor	0.9
Learning rate drop period	400

The learning rate drop factor and period mean that after 400 epochs, the learning rate becomes 0.9 times the original value.

$$\boldsymbol{r} = \sum_{i=1}^{n}(\boldsymbol{X}_i - \frac{\overline{\boldsymbol{X}})(\boldsymbol{Y}_i - \overline{\boldsymbol{Y}})}{\sqrt{\sum_{i=1}^{n}(\boldsymbol{X}_i - \overline{\boldsymbol{X}})^2}\sqrt{\sum_{i=1}^{n}(\boldsymbol{Y}_i - \overline{\boldsymbol{Y}})^2}} \tag{11}$$

When comparing land subsidence prediction performance, RMSE and R^2 are used as comparison metrics, and their calculation formulas are shown in equations (11) –(12), where X represents the true value and Y represents the predicted value.

$$\boldsymbol{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(\boldsymbol{X}_i - \boldsymbol{Y}_i)^2}{\boldsymbol{n}}} \tag{12}$$

$$\boldsymbol{R}^2 = 1 - \frac{\sum(\boldsymbol{X}_i - \boldsymbol{Y}_i)^2}{\sum(\boldsymbol{X}_i - \overline{\boldsymbol{X}})^2} \tag{13}$$

The framework diagram of this article is shown in Fig. 4.

4. Results

4.1. An example showing PCA, NMF and ICA on signal decomposition

We use three simulated signals and synthesize them to verify the capability of PCA, NMF, and ICA in signal decomposition. The simulated signals and their decomposition results are shown in Fig. 5.

From the PCA decomposition results in Fig. 5, it can be observed that simply relying on explaining as much data variance as possible does not achieve good signal decomposition effects, whereas NMF and ICA can relatively well separate mixed signals. Comparing the decomposition results of NMF and ICA, in terms of decomposition accuracy, ICA achieved better performance than NMF, while NMF still contains components from other signals. However, regarding signal decomposition direction, NMF is accurate, whereas the IC1 obtained from ICA decomposition has an opposite direction to the actual signal. A characteristic of ICA is that the directions of the decomposed independent components may change. Although these directional changes are reflected in the coefficients, it is still difficult to determine the direction of coefficients in practical applications, which creates difficulties in identifying the actual settlement features represented by the independent components. Since both the components and coefficients of NMF are non-negative values, no positive–negative cancellation effects occur, making it possible to use NMF to assist in determining the direction of ICA decomposition results.

4.2. Land subsidence monitoring results

Fig. 6 shows the LBP land subsidence rates observed by InSAR (The figure applies water body and low coherence masking, containing only areas with high coherence). The subsidence centers are mainly distributed in the northeastern coastal areas of Dongying and parts of Liaocheng and Dezhou regions, with maximum subsidence rates reaching 275 mm/year. Meanwhile, obvious surface uplift signs appeared in some parts of the region, with maximum uplift rates reaching 90 mm/year, which overall indicates that the groundwater extraction restriction measures promulgated by Shandong Province and artificial measures such as the South-to-North Water Diversion Project have played a good mitigating role in land subsidence in some areas.

4.3. Non-negative matrix factorization results

Before using NMF for decomposing land subsidence time series data, we performed data standardization to ensure that the subsidence data becomes non-negative values, in order to meet the requirements of NMF. The land subsidence time series data in the study area is decomposed using NMF, and the results are shown in Fig. 7. The NMF1 component is a trend-like declining state with clear periodic fluctuations, widely distributed throughout the study area, with the maximum coefficients in

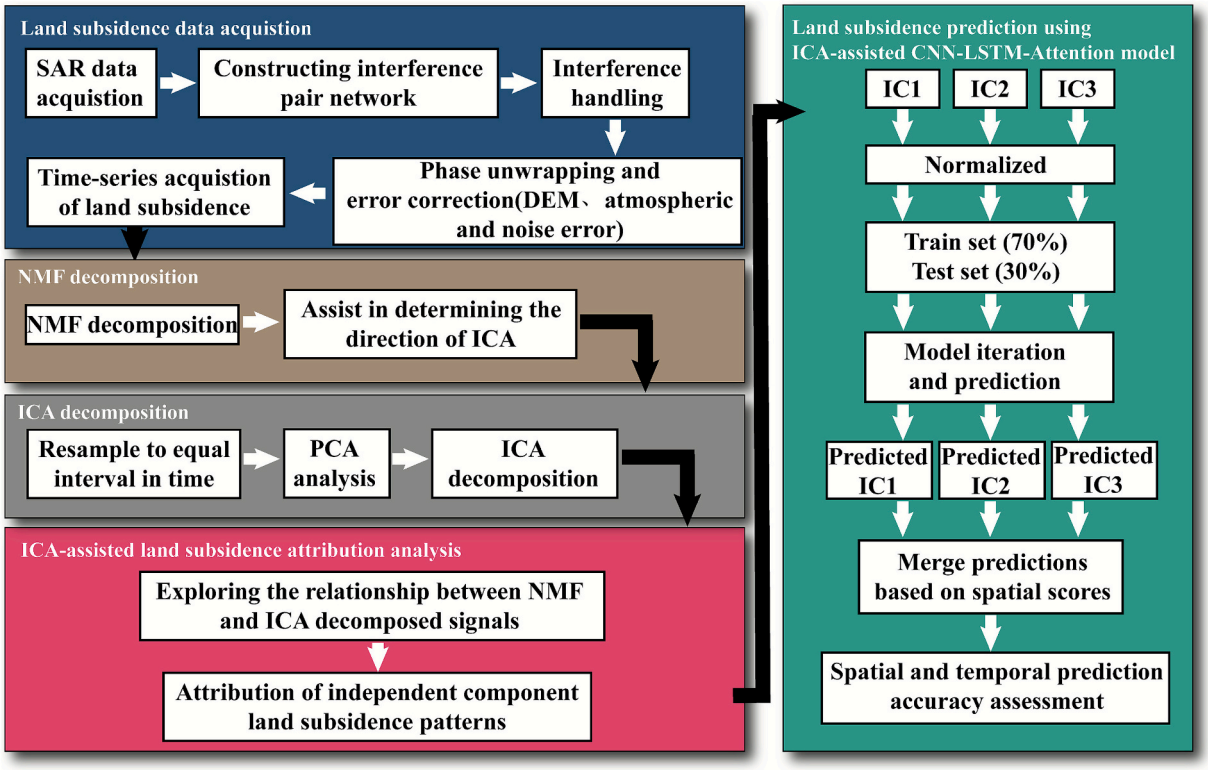

Fig. 4. Process framework diagram of this study.

the eastern coastal and southeastern regions. The NMF2 component is a change pattern with clear periodic fluctuations that shows initial uplift followed by decline, with the maximum coefficients mainly concentrated in the central areas of the study region such as Binzhou and Zibo. The NMF3 component is a slow uplift pattern with clear periodic fluctuations, with the maximum coefficients mainly distributed in small areas of southern Dezhou.

4.4. Independent component analysis results

For the observed land subsidence time series data, we first use PCA for dimensionality reduction to determine the number of independent component decompositions. The first three principal components contain 92.02 % of the information, while the first four principal components contain 94.55 % of the information. Based on the principle of retaining the most information with the fewest principal components, we determine to decompose into three independent components. For the direction of independent components, we refer to the NMF decomposition results for determination.

Fig. 8 shows the three independent components obtained from decomposition and their corresponding spatial scores. IC1 is a near-linear land subsidence pattern that overall exhibits a trend-like declining state with slight fluctuations, mainly occurring in the northeastern coastal and southern areas of Dongying City, as well as parts of Liaocheng and Dezhou. IC2 overall presents a trend of initial rise followed by decline with 2020 as the turning point, mainly occurring in the central regions such as Binzhou and Zibo. IC3 exhibits a subsidence pattern with clear periodicity, mainly occurring in the central-western regions of Liaocheng and Dezhou.

4.5. Regional land subsidence prediction results

We use CNN-LSTM-Attention neural network models to predict each independent component, and the prediction results are shown in Fig. 9. To improve prediction accuracy, we first use cubic functions to fit the trend of each independent component and subtract the trend part from the independent components. A1 and B1 in Fig. 9 show the original independent components, the cubic function fitted trend terms, and the independent components after trend removal. C1 does not require this processing since it has no clear trend characteristics. A2, B2, and C3 are the model prediction results for each independent component, respectively. The first independent component achieves an RMSE of 0.06 and R^2 of 0.86 on the test set after trend removal, the second independent component achieves an RMSE of 0.25 and R^2 of 0.84 on the test set after trend removal, and the third independent component achieves an RMSE of 0.69 and R^2 of 0.80 on the test set. A3 and B3 are the results after adding back the trend terms for the first two independent components.

Furthermore, we obtain the land subsidence prediction results for the entire region by applying the decomposition matrix of the original matrix to the independent components predicted on the test set, as shown in Fig. 10. We selected January, June, and December 2023 from the test set to display the prediction results. We calculated the absolute error between actual subsidence and predicted values, and it can be found that the overall error is highest in the southwestern part in January and June 2023, and higher in the southwestern and northern regions in December 2023. The main reason for this error distribution is the poor prediction accuracy of IC3, leading to high-error zones concentrated in areas with high spatial scores of IC3.

5. Discussion

5.1. The relationship between the signals decomposed by NMF and ICA

For ICA and NMF, both are matrix operation-based signal decomposition algorithms, but the decomposition results of NMF are not independent, meaning that the components are correlated with each other. From the three components observed in Fig. 7, it is found that they all have relatively obvious periodic characteristics. The results of ICA decomposition are mutually independent, and independence necessarily implies no correlation, but no correlation does not necessarily imply

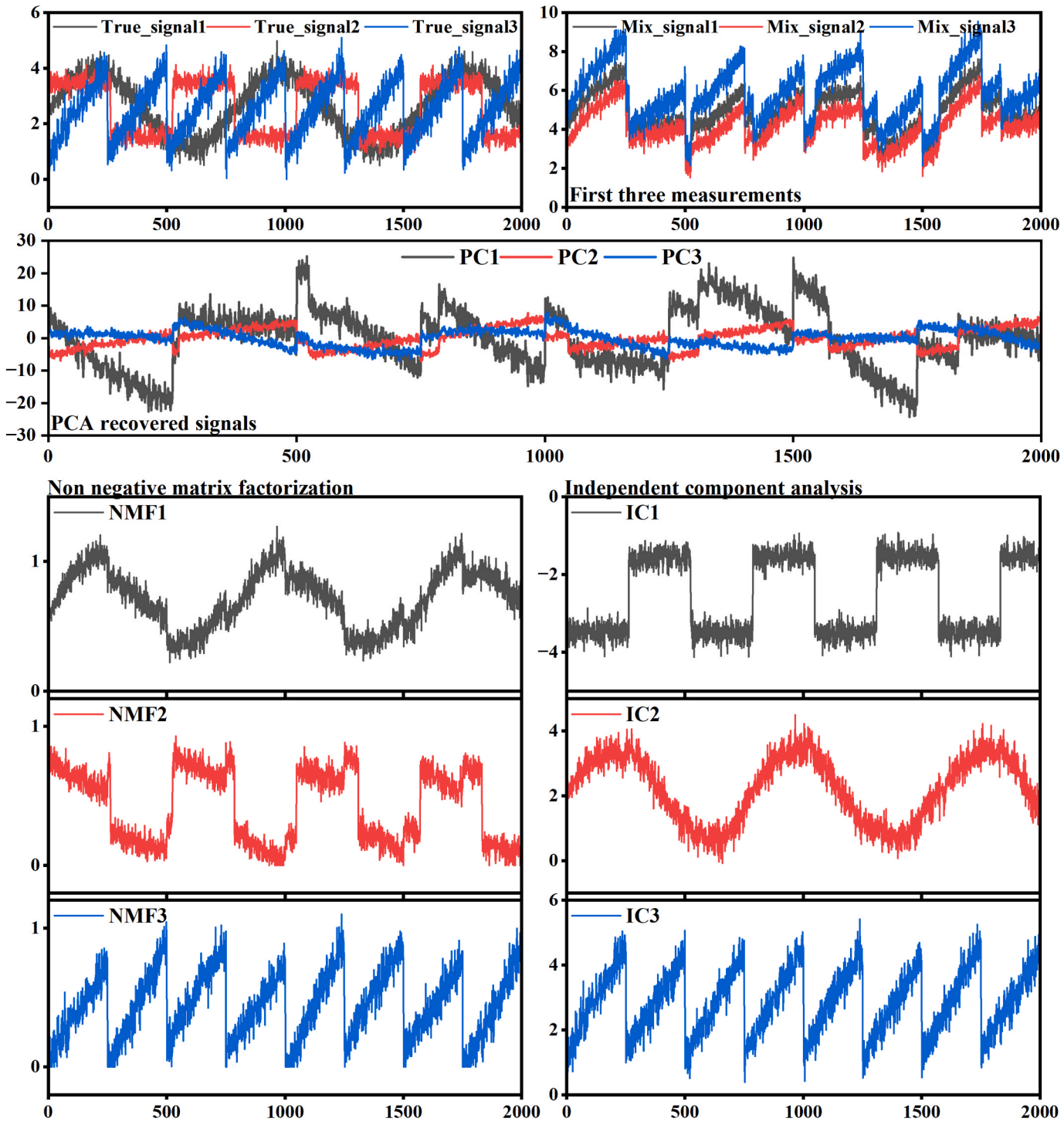

Fig. 5. Schematic diagram of decomposition results of three methods on model signals.

independence (Lee, 1998). We can observe the three independent components from Fig. 8, where only IC3 has relatively obvious periodic characteristics, while the periodic features of IC1 and IC2 are not as clear compared to NMF.

The subsidence sequences extracted by NMF are the three most clear subsidence features in the region. These three features are comprehensive, not mutually independent like independent components, and it can also be said that these three subsidence features are based on independent components as the foundation. NMF can observe the most major subsidence features in the region, while ICA can obtain independent subsidence patterns caused by different driving factors.

First, we observe component1 in Fig. 7, which is a periodic linear subsidence pattern widely distributed throughout the study region, mainly in the eastern coastal areas and western regions such as Dezhou and Liaocheng. Component1 is closely related to the corresponding IC1 and IC3, where IC1 is a linear subsidence pattern and IC3 is an clear periodic subsidence pattern. The high spatial score areas of IC1 are also mainly concentrated in the eastern coastal and western Dezhou regions, while the high spatial score areas of IC3 are concentrated in the southwestern regions such as Dezhou and Liaocheng. However, IC3 has slightly lower spatial scores in the eastern areas, possibly because IC1 has very high spatial scores in these locations, and its own minor

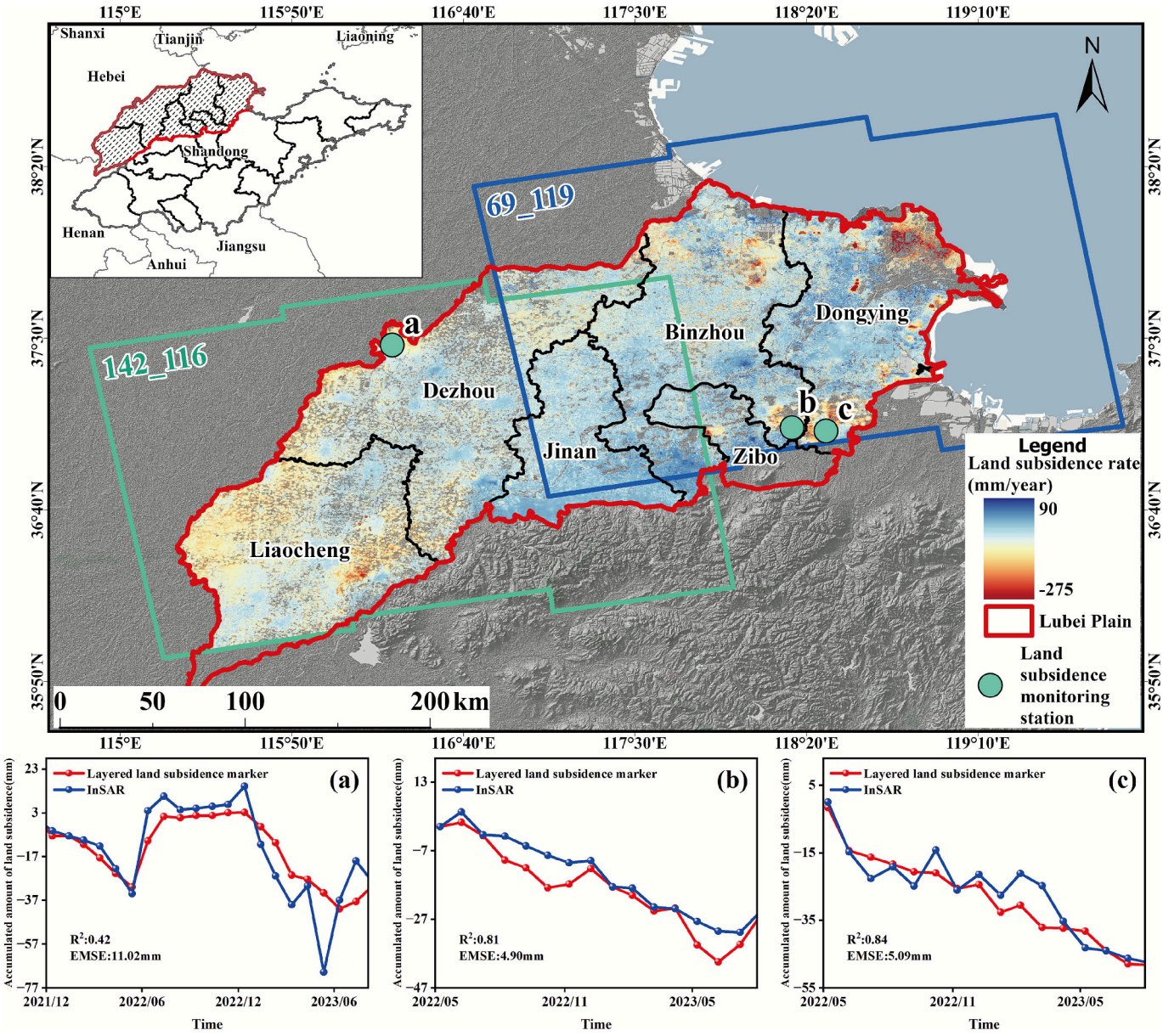

Fig. 6. Land subsidence rates in the study area were obtained using InSAR, and three land subsidence monitoring stations were used to validate the monitoring results. For validation, we selected one InSAR monitoring time point within each month and compared the cumulative subsidence amounts with the corresponding layered benchmark data at the same time point.

fluctuation changes are amplified into more clear periodic changes.

Secondly, for component2, this is a change pattern with periodic fluctuations that shows initial rise followed by decline, with larger coefficients concentrated in the central regions of the study area near Binzhou and Zibo. Component2 is closely related to IC2 and IC3, where IC2 is a trend-like change that first rises then falls, and its high spatial score areas are also concentrated near central Binzhou and Zibo. Although IC3′s spatial scores in this area are not as high as in the southwestern regions, they are relatively higher compared to coastal areas. Meanwhile, due to IC2′s high spatial scores in this location, some weak periodic features of IC2 itself are also amplified.

Finally, component3 is a slow rebound pattern with periodic fluctuations, with correspondingly low spatial coefficients overall, with relatively higher values concentrated in a small area in southern Dezhou. IC3 has the greatest relationship with it, as its spatial scores are higher in this region. In summary, this demonstrates that the NMF decomposition results are closely related to the ICA decomposition results, with the NMF decomposition results resembling combinations based on independent components as the foundation.

5.2. Attribution of land subsidence patterns represented by independent components

The Land cover conditions in the study area from 2018 to 2023 are shown in supplementary materials Fig. S1, where it can be found that the primary Land cover type in the region is cultivated land, followed by red impervious surfaces, which represent urban areas. It can be observed that cultivated land is most widely distributed in western Dezhou and Liaocheng, while other cities have relatively more diverse Land cover types.

Agricultural irrigation is another important factor affecting land subsidence. We utilized farmland irrigation water consumption data from water resources bulletins as shown in supplementary materials Fig. S2. It can be found that the farmland irrigation water consumption

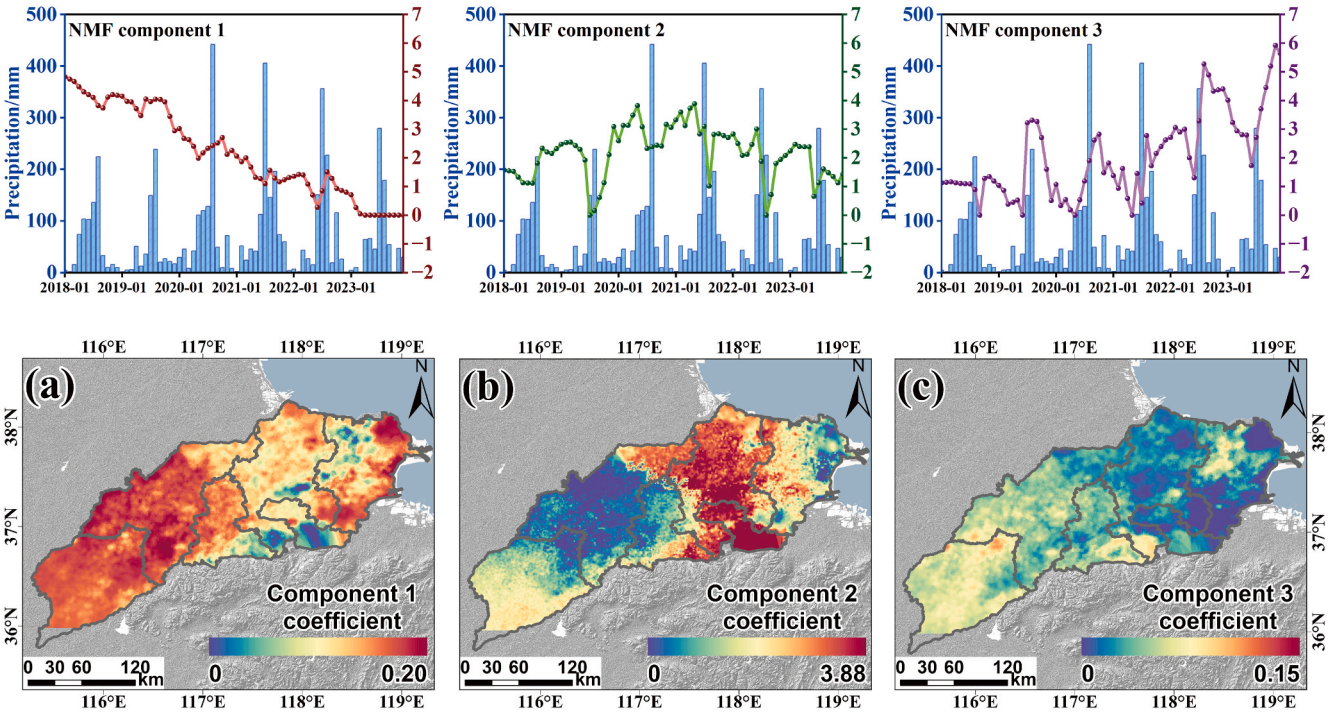

Fig. 7. The three components obtained by NMF decomposition and their corresponding coefficients in space.

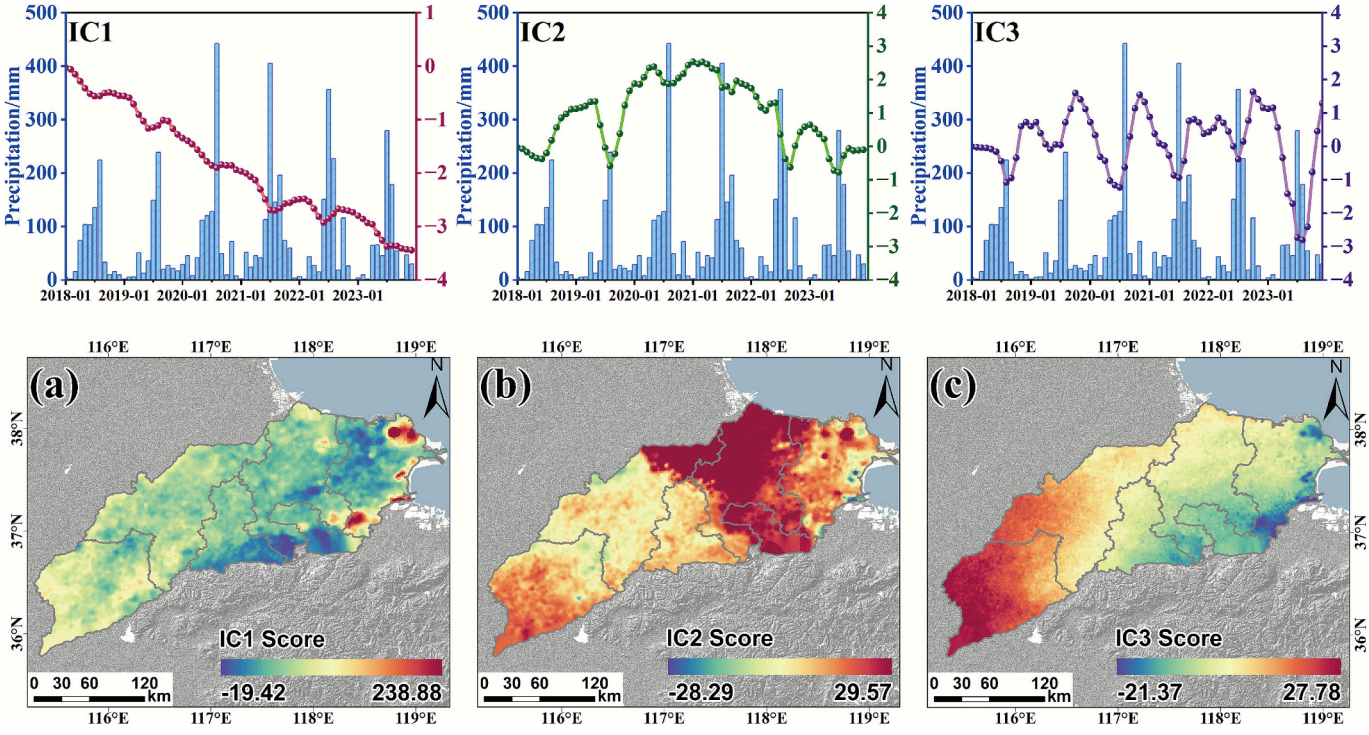

Fig. 8. Three independent components obtained by ICA decomposition and their corresponding spatial scores.

is consistent with the Land cover data. As shown in Fig. S3 of the supplementary materials, from the perspective of multi-year average precipitation, areas with higher precipitation occur in the southeastern region, showing an overall decreasing trend from southeast to northwest.

5.2.1. Attribution analysis of the land subsidence pattern corresponding to IC1

For the land subsidence pattern represented by IC1, we selected 18 groundwater observation wells within the high-value areas of its spatial scores, including 4 confined water observation wells and 14 unconfined water observation wells. Comparative analysis was conducted between the actual land subsidence observation data and water level monitoring data at the observation wells and IC1, with results shown in Fig. 11. We

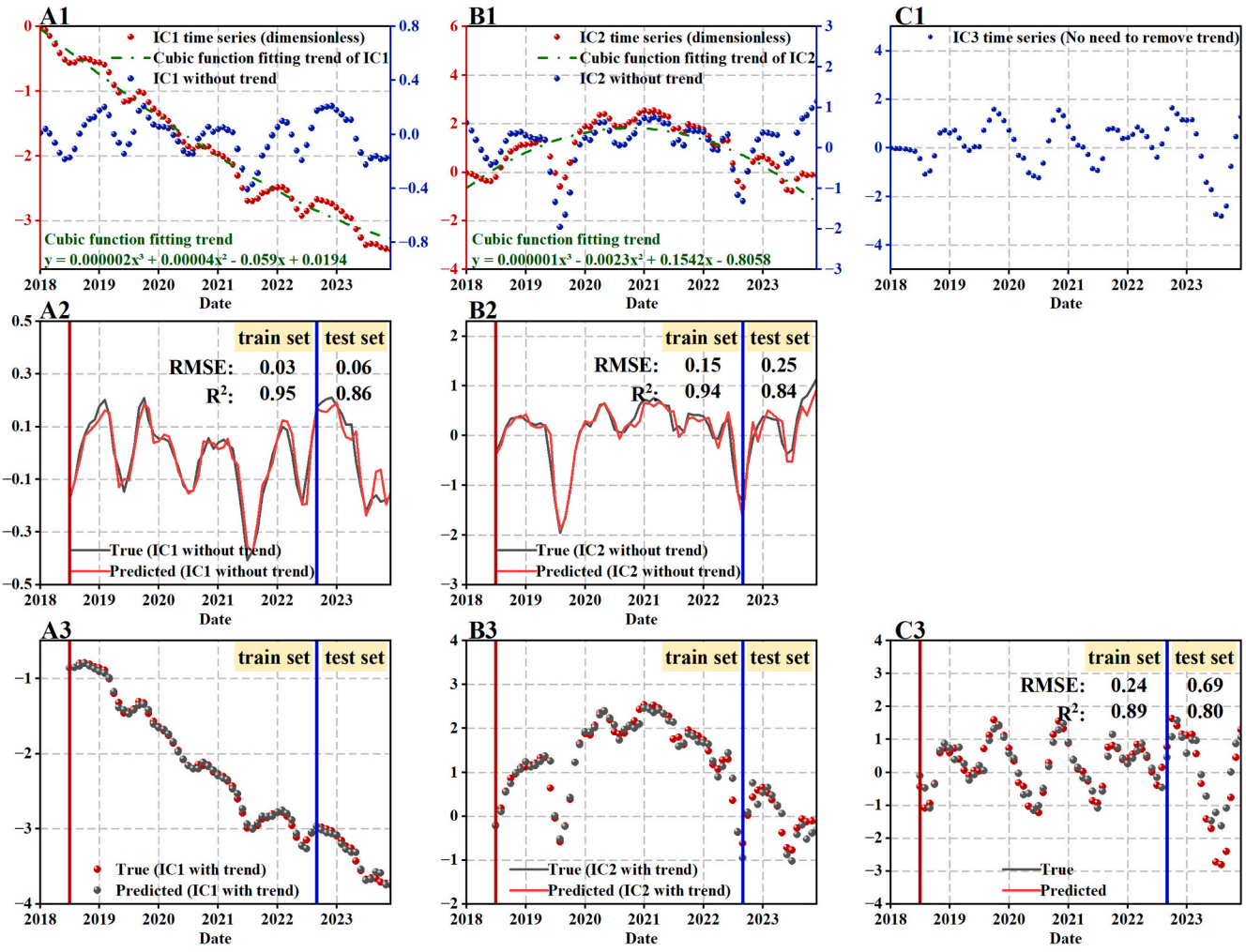

Fig. 9. Prediction results of the CNN-LSTM-Attention neural network model on each independent component.

can find that IC1 well describes the regional land subsidence characteristics. By calculating correlation coefficients, it is found that the correlation coefficients at the 18 observation wells range from a minimum of 0.80 to a maximum of 0.97.

However, there is an interesting phenomenon here where GWL show clear rebound while land subsidence continues to occur, a phenomenon that has been observed throughout the Lubei Plain (Wang et al., 2024). One possible reason for this phenomenon is the inconsistency between the aquifers from which groundwater is extracted in this region and the aquifers monitored by observation wells. The main groundwater utilization methods in this region include extracting brine for salt production (Wang et al., 2022; Zhang et al., 2021), some farmland irrigation and extracting groundwater for aquaculture (Higgins et al., 2013), among others. Different water use methods extract from different aquifers, which subsequently leads to different changes in different compressible layers, resulting in low correlation between GWL and independent components.

Another reason is caused by petroleum extraction in the region (Jing et al., 2021; Künzer et al., 2014), where long-term high-intensity over-exploitation leads to reservoir pressure decline, compression of clay layers and oil and gas fields, which subsequently causes land subsidence (Feng and Xue, 2004). Although subsidence can be mitigated by injecting water into oil reservoirs after extraction, the shallow saltwater extracted for injection simultaneously produces greater subsidence (Zhang et al., 2014). Finally, groundwater observation wells, as point measurements, have certain limitations when evaluating regional land subsidence.

To further study the Land cover conditions of subsidence centers, we used Sentinel-2 data to obtain true color images with 10 m spatial resolution of this region, as shown in Fig. 12. From Fig. 12, in the high-value areas of spatial scores for A, B, C and their local enlarged maps, it was found that the main land cover types in these areas include farmland, salt fields, and aquaculture.

In the high-value areas of spatial scores for D and its locally enlarged section in Fig. 12, a large petroleum processing enterprise is found. Petroleum companies conducting petroleum processing activities utilize groundwater resources, leading to subsidence occurrence. Land subsidence caused by petroleum extraction is still occurring (Lu et al., 2024), so the near-linear land subsidence pattern appearing in this region is mainly driven by factors that may include direct oil extraction or the extraction of deep groundwater for petroleum processing.

In the high-value areas of spatial scores for E, F, G and their locally enlarged sections, it was found that the main land use type is farmland. These areas are concentrated in Dezhou, Liaocheng and other regions, mainly due to large-scale extraction of deep confined water for irrigation causing trend-related land subsidence (Wang et al., 2024; Xu et al., 2025). Therefore, the linear subsidence pattern corresponding to IC1 is mainly caused by over-exploitation of deep confined water for farmland irrigation, brine extraction for aquaculture or salt production, and petroleum extraction and processing, corresponding to deformation of deep compressible layers.

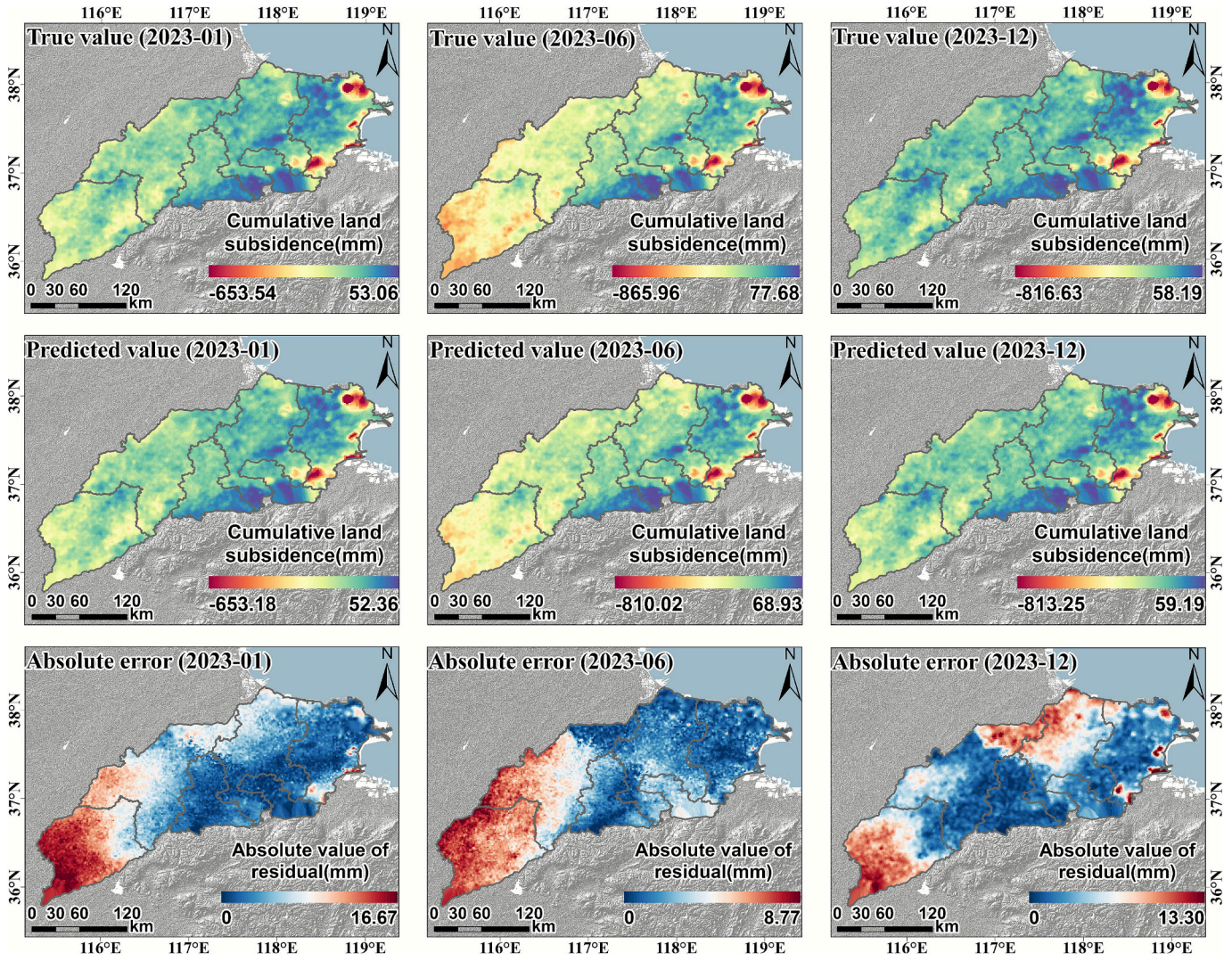

Fig. 10. Land subsidence prediction results and corresponding absolute errors in the entire study area.

5.2.2. Attribution analysis of the land subsidence pattern corresponding to IC2

For the land subsidence pattern represented by IC2, we selected 14 groundwater observation wells within the high-value areas of its spatial scores, including 8 confined water observation wells and 6 unconfined water observation wells. Comparative analysis was conducted between the actual land subsidence observation data and water level monitoring data at the observation wells and IC2, with results shown in Fig. 13. It can be found that IC2 also well describes the land subsidence pattern in this region, with the highest correlation coefficient being 0.72. Unlike the phenomenon observed with IC1, GWL at observation wells in high spatial score areas of IC2 also show good correlation with the independent component, with the maximum correlation coefficient being 0.56.

We observe the subsidence pattern represented by IC2 in Fig. 8 and find that it has good consistency with interannual precipitation changes, with 2020 having the highest precipitation, and the corresponding independent component also reaching its maximum value. Therefore, IC2 may represent a subsidence pattern caused by interannual precipitation changes, but the high spatial score areas of this pattern do not completely correspond with areas of higher precipitation. The reason for higher spatial scores in Binzhou and southern Zibo is that other regions have clear trend-like subsidence, and this precipitation-induced subsidence pattern is relatively masked by linear rapid subsidence characteristics, making it more difficult during independent component decomposition. Since IC2 does not show changes related to precipitation cycles, although we attribute IC2 to precipitation-related factors, this correlation is related to precipitation total trends. Deep compressible layers show IC1-type linear trend changes, while surface deformation should show some relationship with precipitation cycles, so we believe this is deformation of middle-shallow compressible layers.

5.2.3. Attribution analysis of the land subsidence pattern corresponding to IC3

For the land subsidence pattern represented by IC3, we selected 17 unconfined and 1 confined water observation wells within the high-value areas of its spatial scores. Comparative analysis was conducted between the actual land subsidence observation data and water level monitoring data at the observation wells and IC3, with results shown in Fig. 14. Similar to IC2, IC3 shows good correlation with both actual land subsidence observation data and water level observation data from groundwater observation wells, with the highest correlation coefficient with land subsidence observation data reaching 0.63, and the highest correlation coefficient with water level observation data reaching 0.71.

The independent component time series obtained from independent component analysis are dimensionless values, and the positive or negative signs have no actual meaning and can be changed by multiplying by −1. However, this is an interesting issue for periodic

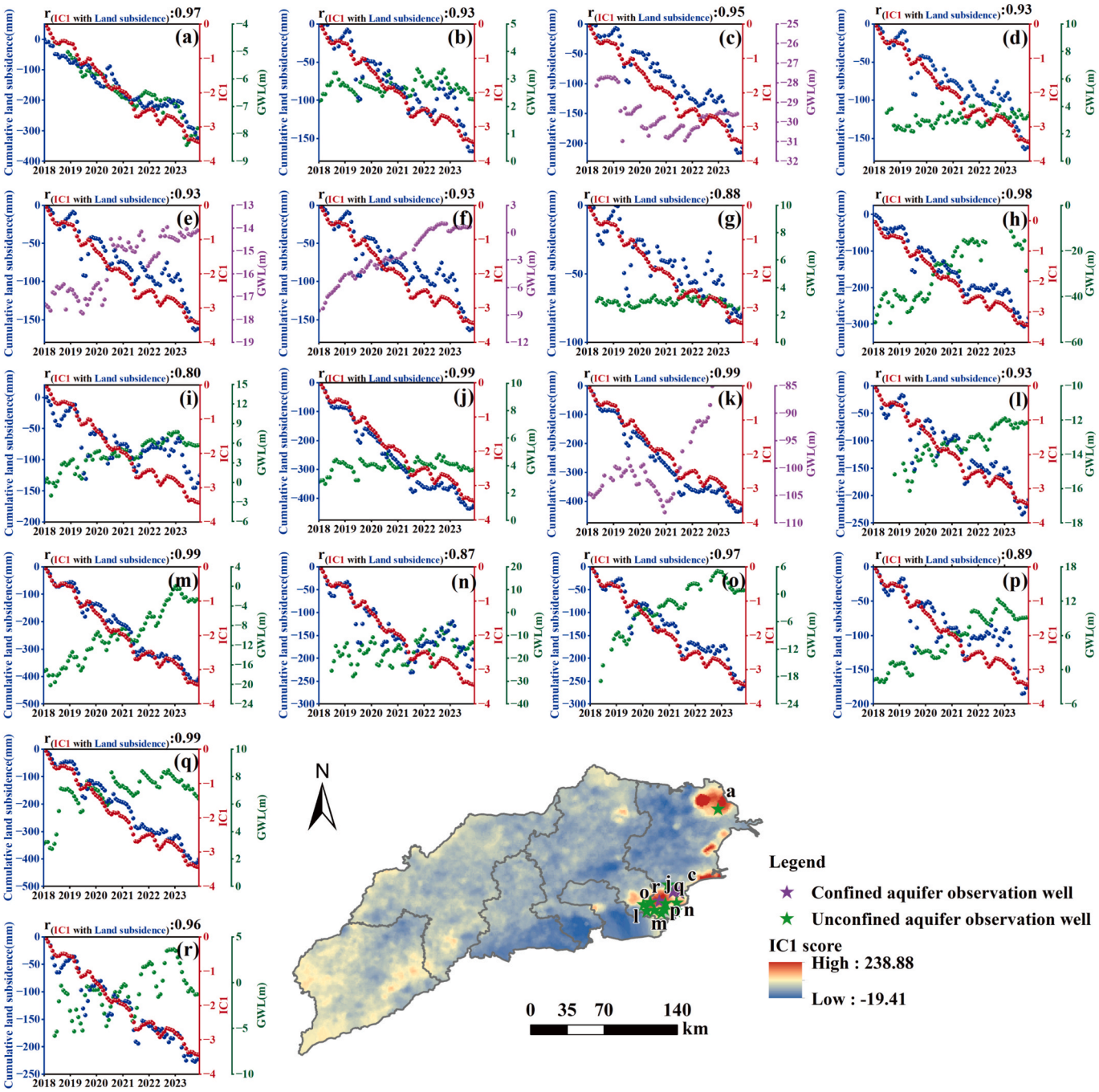

Fig. 11. The locations of groundwater observation wells selected in the IC1 high-score area and the IC1, GWL and land subsidence accumulation data were plotted.

independent components, because once periodicity is reversed, two phase-opposite periodic signals will appear. In the study region, the main driving factors that can cause periodic land subsidence include seasonal agricultural irrigation and periodic precipitation. Fig. 15 shows the independent component IC3 before and after reversal and its corresponding spatial scores.

We can find that the independent component time series corresponding to Fig. 15 (a) has an clear lag period with precipitation, with peaks mostly occurring in autumn and winter, and troughs appearing in spring and summer. Its high spatial score areas are concentrated in regions such as Liaocheng and Dezhou. Combined with the agricultural irrigation water consumption data shown in supplementary materials Fig. S2, it can be found that this region has large water consumption for agricultural irrigation. Combined with the characteristics of crops planted in the region, winter wheat is planted from October each year to June of the following year, with the first irrigation around November when irrigation volume is relatively small. The second irrigation period is around February of the following year, after which large amounts of groundwater need to be extracted for irrigation until harvest. After harvest, summer corn needs to be planted around June and harvested around September. Precipitation from June to September is the highest throughout the year, summer corn requires less water resources than winter wheat, and with natural precipitation supplementation, land subsidence is alleviated (Song, 2023; Wang et al., 2007). Moreover, combined with the spatial distribution map of multi-year average precipitation in supplementary materials Fig. S3, precipitation in this region is relatively less compared to eastern areas. Comprehensive consideration shows that the main driving factor causing periodic land

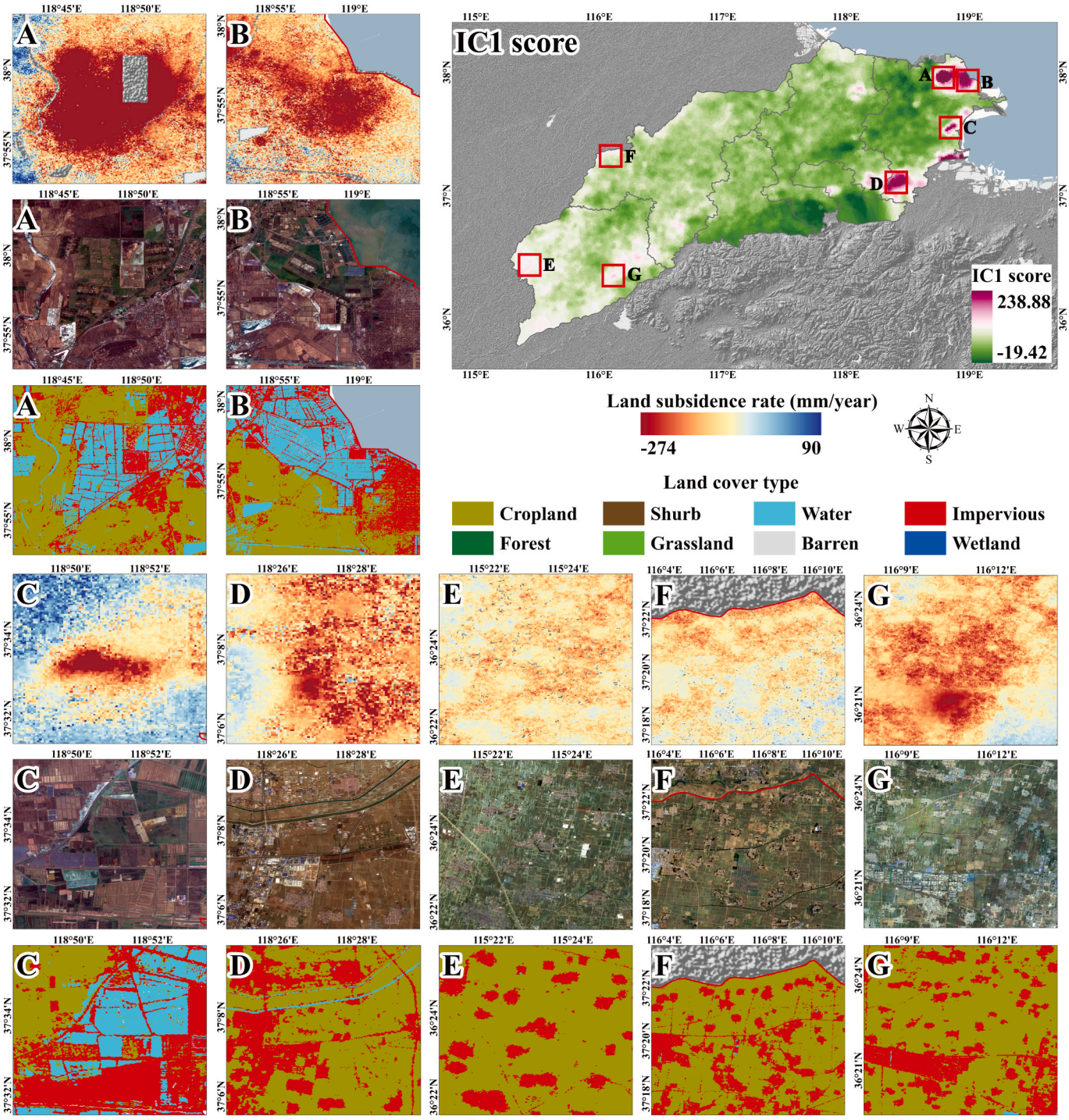

Fig. 12. Land subsidence rates and land use status in representative areas with high IC1 spatial scores (compared using Sentinel-2 and land cover data).

subsidence in this region is seasonal agricultural irrigation, and the compressible layers producing this subsidence characteristic should still be middle-shallow layer changes. The reason is that it has a relatively long lag time compared to periodic precipitation changes, deep compressible layers are experiencing IC1-type linear subsidence, and the burial depth of middle-shallow layers also corresponds to the lag of water level changes or compressible layer changes relative to irrigation water sources.

Meanwhile, for the reversed independent component shown in Fig. 15 (b), its peaks mostly occur in spring and summer, which has high correlation with precipitation. Its high spatial score areas are mostly concentrated in the southeastern region. Combined with the precipitation spatial distribution map shown in supplementary materials Fig. S3, precipitation in this region is relatively higher compared to the northwest, and combined with the Land cover data in supplementary materials Fig. S1, it is found that urban land and other non-agricultural Land covers in this region are relatively more abundant compared to the central-western parts, while the farmland irrigation water consumption shown in supplementary materials Fig. S2 is also less. Therefore, comprehensive analysis shows that the main driving factor for periodic land subsidence occurrence in this region is seasonal precipitation, mainly representing surface micro-deformation driven by precipitation. Many authors attribute the driving factors of periodic independent components to precipitation and seasonal agricultural irrigation when

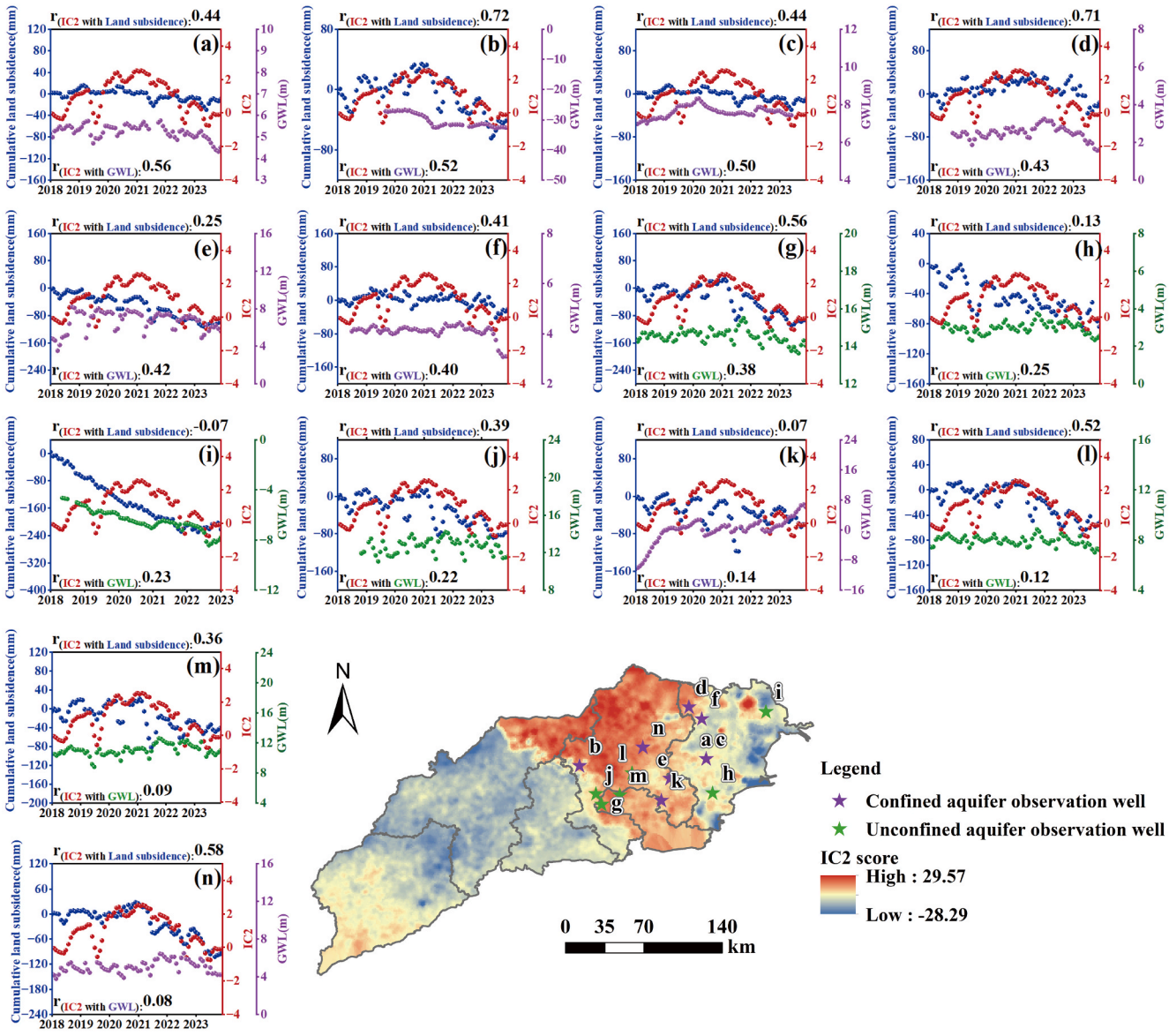

Fig. 13. The locations of groundwater observation wells selected in the IC2 high-score area and the IC2, GWL and land subsidence accumulation data were plotted.

conducting analysis (Song, 2023; Wang et al., 2025), but do not conduct further decomposition and attribution of the two factors. This study provides a new interpretive approach to help further analyze the driving factors of subsidence occurrence.

5.3. *The role of CNN-LSTM-Attention combined with ICA in land subsidence prediction*

When performing subsidence time series prediction, we first removed trend terms and obtained cubic fitting equations for the trend terms, so future trend terms can be obtained through the equations. However, it should be noted that the assumption for using this method is that the overall change trend of land subsidence time series will not undergo particularly dramatic changes, so this fitted trend is accurate in the short term. Secondly, we predict the detrended time series without using other covariate features, which may produce error accumulation during long-term prediction (Khashei et al., 2009), but for short-term future prediction capability, as shown in our validation results, relatively ideal results can be achieved.

Similarly, the assumption that land subsidence time series change trends do not undergo dramatic changes also ensures consistency of spatial scores in the short term, which provides assurance for reconstructing independent components into land subsidence across the entire space through decomposition source matrices. One advantage of this method is that it eliminates the need for individual prediction of land subsidence time series for each point in space, thereby reducing the workload and achieving regional land subsidence prediction through this approach. Another advantage is that PCA dimensionality reduction is performed before conducting independent component analysis, which removes some noise and very weak variation patterns, retaining only the most significant land subsidence characteristics, thereby reducing the complexity of time series and improving prediction accuracy.

5.4. *Limitations and shortcomings*

(1)We use SBAS-InSAR to obtain land subsidence data for the study area, which has relatively lower reliability compared to subsidence data obtained by PS-InSAR, but SBAS-InSAR can obtain more subsidence point information. Attempting to improve subsidence data acquisition methods while enhancing both monitoring accuracy and monitoring coverage is the next research direction.

(2)When performing ICA decomposition, different compressible

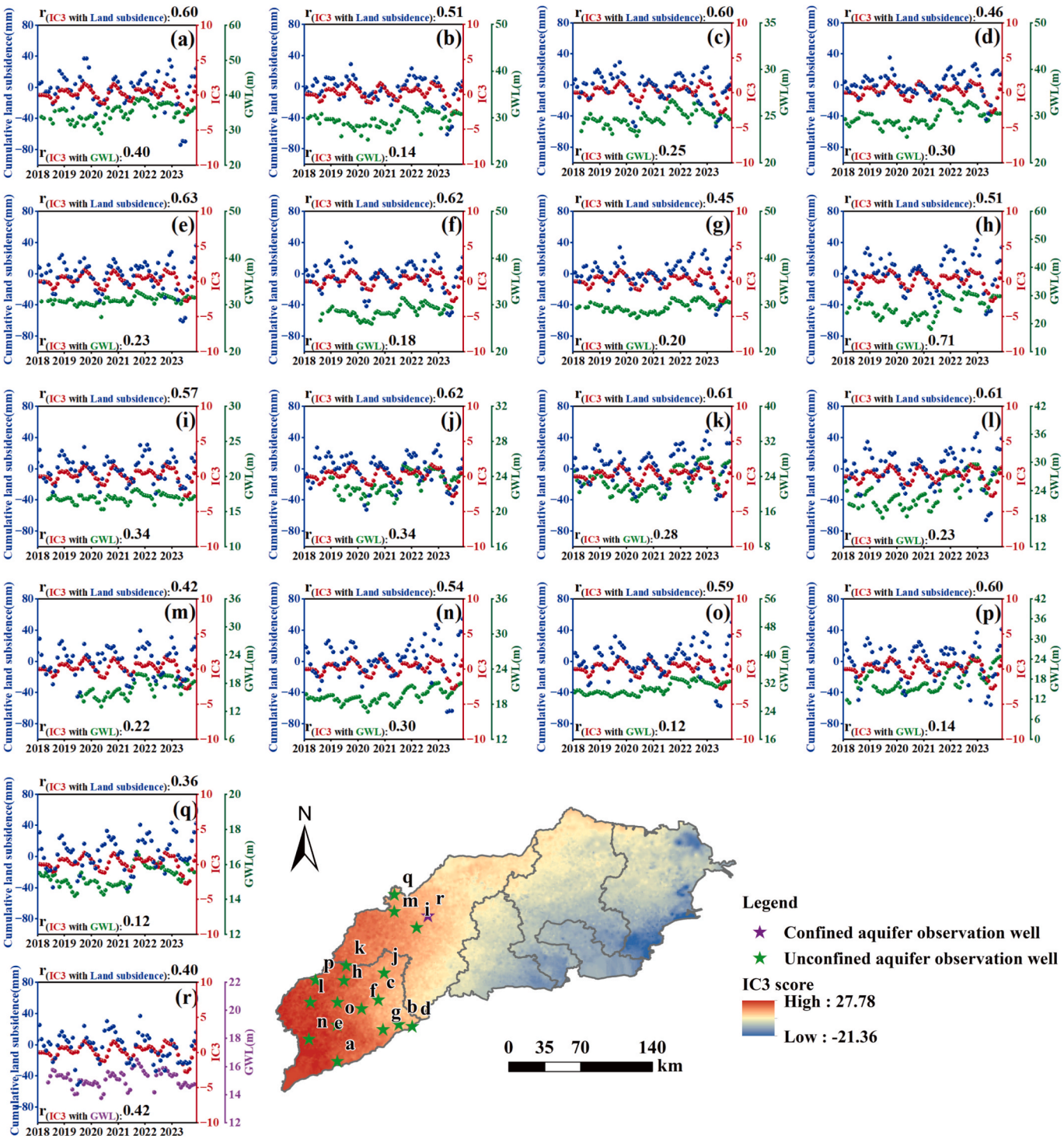

Fig. 14. The locations of groundwater observation wells selected in the IC3 high-score area and the IC3, GWL and land subsidence accumulation data were plotted.

layers cannot achieve complete independence, and hydraulic connections across aquifers may exhibit similar subsidence characteristics. Therefore, for ICA that achieves decomposition based on non-Gaussian independence, the independent components obtained from decomposition require considerable a posteriori information for attribution, which brings some challenges to identifying the driving factors that trigger subsidence.

(3) Previous research successfully decomposed different land subsidence characteristics caused by sandy soil and clay in LBP. We focus more on the decomposition of driving factors and burial depths of different land subsidence patterns (Su et al., 2025). Further decomposition based on burial depth and soil properties is of great significance for understanding the characteristics of LBP land subsidence. It can also provide a better understanding of the attribution of existing land subsidence driving factors and make up for existing deficiencies.

6. Conclusions

This study uses Non-negative matrix factorization (NMF) assisted independent component analysis to decompose the characteristics of

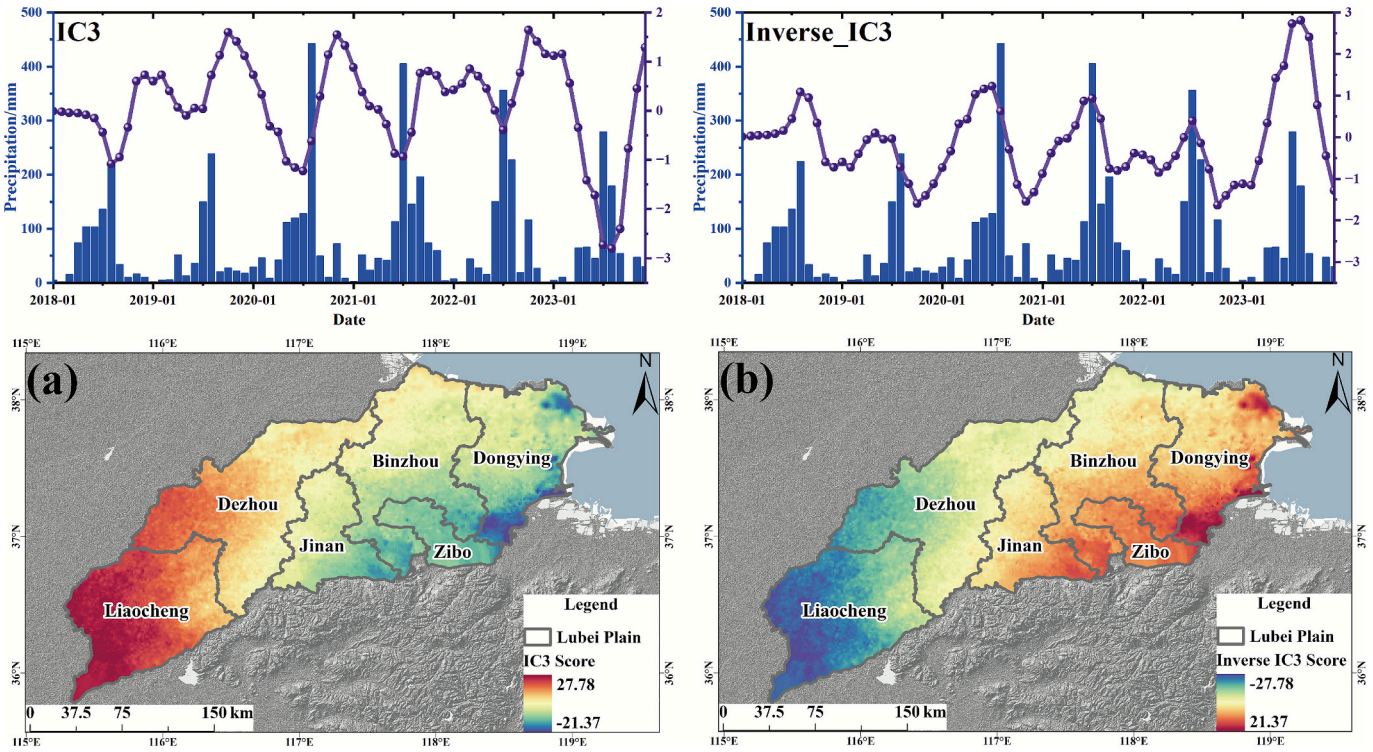

Fig. 15. The new independent components obtained after reversing IC3 and their spatial scores.

land subsidence data in the Lubei Plain from 2018 to 2023, and attributes the driving factors of the subsidence patterns corresponding to the three independent components obtained from decomposition. Then, CNN-LSTM-Attention neural networks are used to predict each independent component separately, and the land subsidence prediction results for the entire region are reconstructed through decomposition source matrices. Key contributions include: (1) The three components obtained from NMF have close correlations with the decomposed independent components. Although they cannot accurately decompose each signal like ICA, the decomposition directions are correct, assisting in determining the decomposition direction of each independent component. (2) IC1 achieves correlation coefficients above 0.8 with actual land subsidence observation data at 18 groundwater observation wells within its high spatial score areas, but GWL show an uplift pattern opposite to subsidence. IC1 represents a near-linear subsidence pattern triggered by long-term extraction of deep confined water for farmland irrigation, aquaculture, brine salt production, and petroleum extraction, mainly corresponding to deformation of deep compressible layers. IC2 achieves correlation coefficients up to 0.72 with actual land subsidence observation data and up to 0.56 with water levels at 14 groundwater observation wells in its high spatial score areas. IC2 represents a change pattern with 2020 as the turning point, showing initial rebound followed by subsidence. This pattern is closely related to interannual trend changes in total precipitation and mainly corresponds to deformation of middle-shallow compressible layers. IC3 achieves correlation coefficients up to 0.63 with actual land subsidence observation data and up to 0.71 with water levels at 18 groundwater observation wells within its high spatial score areas. IC3 represents periodic subsidence patterns triggered by seasonal farmland irrigation and precipitation. Seasonal farmland irrigation mainly corresponds to deformation of middle-shallow compressible layers, while precipitation-induced periodic deformation mainly corresponds to surface micro-deformation. (3) The proposed subsidence prediction model demonstrates good performance with RMSE ranging from 0.06 to 0.69 and R^2 from 0.80 to 0.86 in test set predictions. This study provides insights for driving factor attribution of regional land subsidence and regional groundwater utilization, and offers a new approach for regional land subsidence prediction.

CRediT authorship contribution statement

Yongkang Wang: Writing – original draft, Software, Methodology. **Huili Gong:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Beibei Chen:** Supervision, Software, Methodology, Conceptualization. **Chaofan Zhou:** Supervision, Methodology, Conceptualization. **Lin guo:** Supervision, Software, Conceptualization. **Haotong Wang:** Supervision, Conceptualization. **Qin Wang:** Supervision, Conceptualization. **Shuqi Hao:** Methodology, Conceptualization. **Yabin Yang:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jag.2026.105097.

Data availability

The authors do not have permission to share data.

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